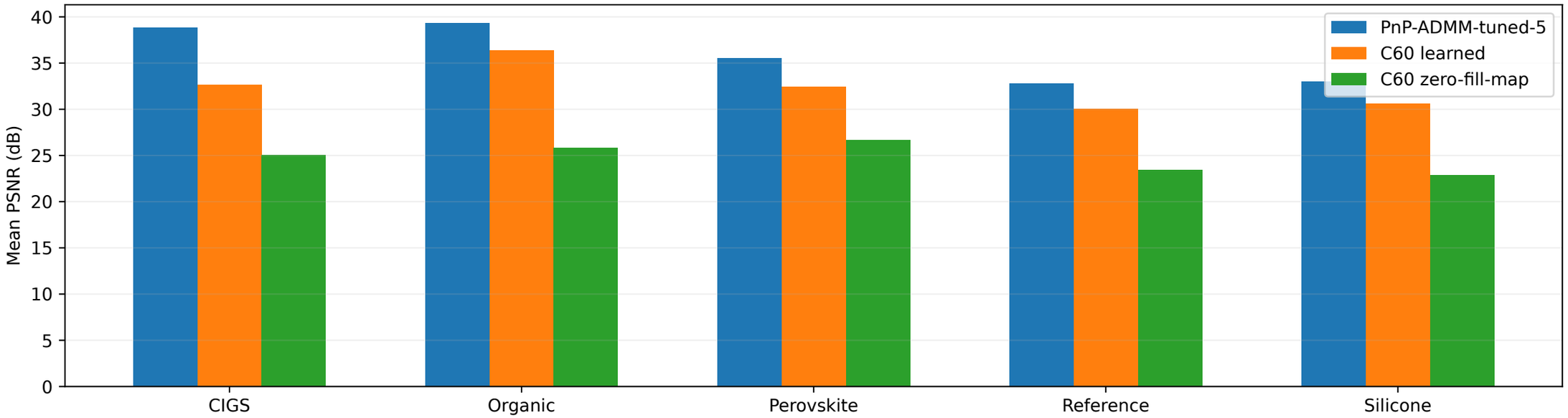
C60 regularisation: learned map vs zero-fill map

Executive summary | 10 frozen validation images (2/category), 20% sampling, NPL-default noise

Average PSNR (dB): PnP-ADMM-tuned-5 35.896 | C60 learned 32.427 | C60 zero-fill-map tuned 24.790

Zero-fill map beats learned map on 0/10 cases. Best rho at lower search boundary on 9/10 cases; best observed rho lies outside final ternary bracket on 3/10.

Each zero-filled parameter map is nonnegative: $\Lambda_\rho = \rho \max(x_{\mathrm{ZF}}, 0)$, because raw zero-fill contains negative values.



Interpretation

The replacement map uses the measured image directly, not a trained CNN. Its absolute regularisation scale and spatial distribution differ from the learned map. The multiplier is chosen with ground-truth PSNR for each image, so these are oracle results. This is a validation subset also used in earlier model/hyperparameter selection, not an independent estimate of generalisation.

Per-case methods and images follow; every reconstruction uses the identical archived acquisition for that case.

Per-image PSNR and selected multipliers

A positive delta means the zero-fill-map reconstruction outperforms learned-map C60.

Image	Selected rho	PnP PSNR	Learned PSNR	ZF-map PSNR	Delta (dB)	Outside bracket
CIGS_0768	0.0001	37.394	32.216	24.335	-7.881	No
CIGS_0158	0.0001	40.237	33.044	25.764	-7.280	No
Organic_0917	0.0001	39.558	36.502	25.913	-10.589	Yes
Organic_0008	0.0001	39.126	36.308	25.773	-10.534	Yes
Perovskite_0733	0.0001	38.556	36.290	28.706	-7.584	No
Perovskite_0396	0.0001	32.517	28.555	24.706	-3.849	No
Reference_0768	0.0001	33.480	31.750	25.085	-6.665	No
Reference_0589	0.0001	32.073	28.389	21.814	-6.575	No
Silicone_0008	0.001	31.869	29.791	23.249	-6.542	Yes
Silicone_0589	0.0001	34.146	31.428	22.552	-8.876	No

Search protocol

Linear-rho ternary search on [0.0001, 10], stop when bracket width <= 0.01. Additional log-spaced diagnostics and final bracket probes are evaluated.

Select best observed PSNR over all evaluated candidates. Ternary search is not a global guarantee without unimodality; boundary and out-of-bracket selections are flagged.

Fixed C60 checkpoint epoch 60, 20 CG input iterations, 512 PDHG iterations. PnP uses the existing validation-selected rho=0.5, sigma=0.075 and residual stopping rule.

Measurements, indices, PnP outputs and both UTV variants are frozen per image and downloadable as NPZ.

Methods, positivity and comparability

One changed ingredient: the spatial regularisation weight map supplied to the unchanged C60 solver.

Learned C60 baseline

$\lambda_{\text{learned}}(x) = \text{softplus}(\text{CNN}(x_{\text{CG}}))(x)$, predicted once from the 20-step CG image.
Its weights are not rescaled.

Replacement map

$\lambda_{\text{ZF},\rho}(x) = \rho \max(x_{\text{ZF}}(x), 0)$, with no CNN evaluation, no map normalisation, and no upper clipping. Negative zero-fill pixel values are set to zero because L1 penalties require nonnegative weights.

Fixed forward data

For each selected case, all reconstructions use the same archived 20% multilevel Hadamard measurement, NPL-default noise, sampling indices and GT.

Fixed reconstruction

C60 TV-PDHG uses the same checkpoint, 512 iterations, zero initial image, step sizes and normalized PCM operator. Only lambda_reg differs.

PnP baseline

PnP-ADMM-tuned-5 uses validation-selected (rho=0.5, sigma=0.075), the original DRUNet model and residual-based stopping, without per-image GT tuning.

Rho tuning

Each replacement map is multiplied by one scalar rho. Ternary stopping width 0.01 refers to rho, not PSNR accuracy. Best observed rho may lie at search boundaries.

Interpretation limit

The C60 model and PnP hyperparameters have previously been selected on validation data. Also, the new rho is explicitly tuned using GT on these validation cases. Results are exploratory, not held-out test results.

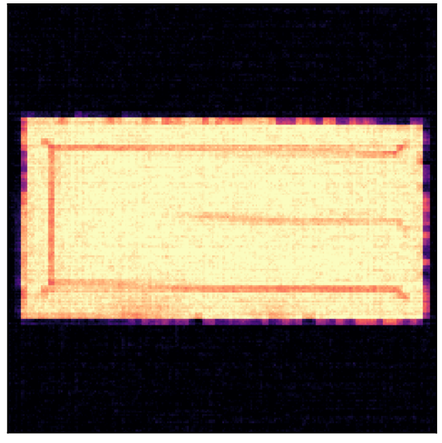
01. CIGS_0768

Frozen validation position 907 | 512 PDHG iterations | PSNR-optimal evaluated rho=0.0001

Ground truth



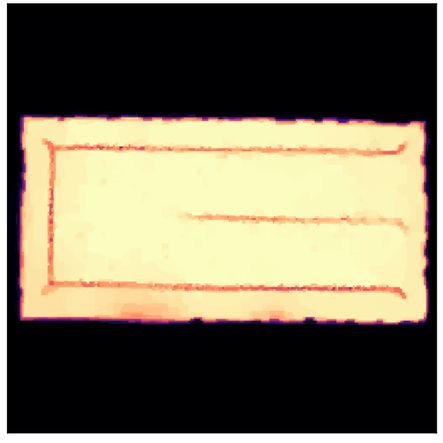
Zero-filled reconstruction
PSNR 24.19 dB | SSIM 0.436
HaarPSI 0.460



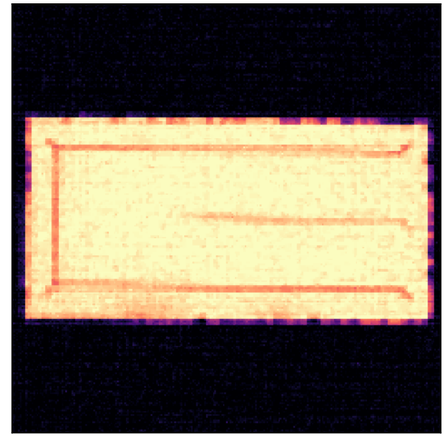
PnP-ADMM-tuned-5
PSNR 37.39 dB | SSIM 0.984
HaarPSI 0.943



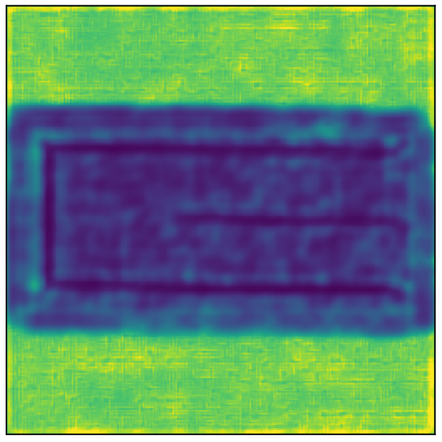
C60 with learned map
PSNR 32.22 dB | SSIM 0.954
HaarPSI 0.869



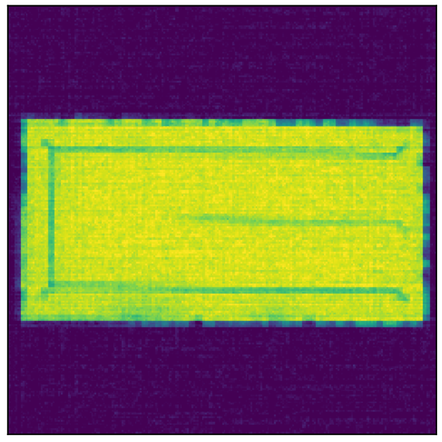
C60 with ZF map (rho=0.0001)
PSNR 24.34 dB | SSIM 0.450
HaarPSI 0.482



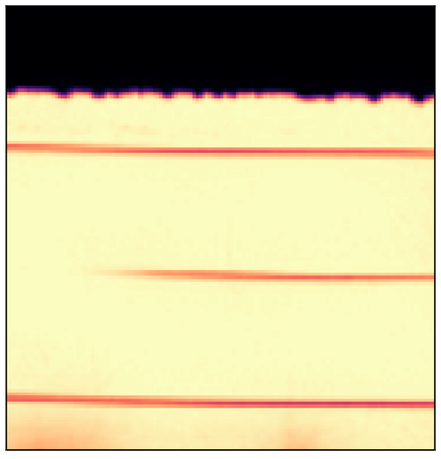
Learned parameter map
min 0.00399 | mean 0.113 | max 0.286



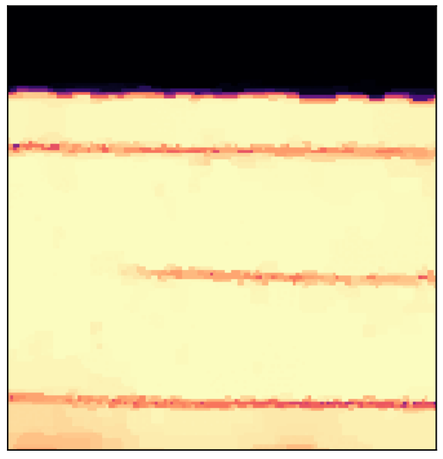
Zero-fill parameter map * rho
min 0 | mean 4.33e-05 | max 0.000113



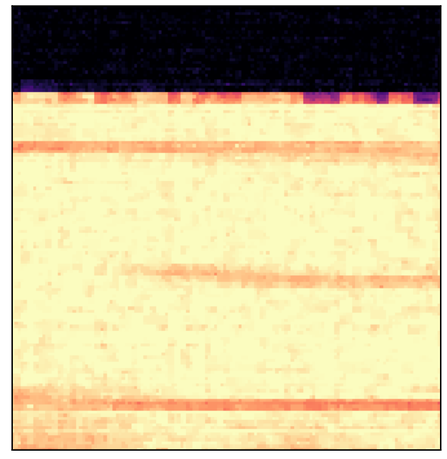
GT texture crop



Learned-map C60 crop



Zero-map C60 crop



PSNR change ZF-map minus learned: -7.881 dB | rho evaluations: 50 | final bracket width: 0.00677

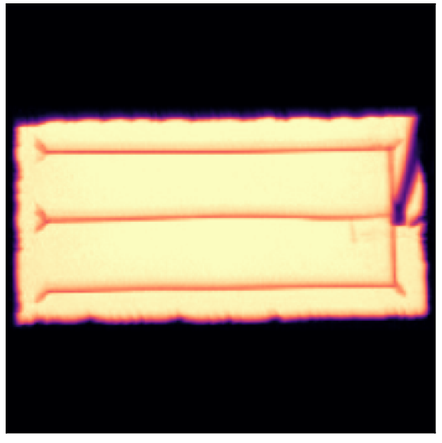
Zero-fill negative pixel fraction: 26.3%. Best rho inside final bracket. Map panels have independent display scales; image panels all use [0,1].

Single-image GT-guided PSNR search. High-quality outputs and search traces are retained in the experiment archive.

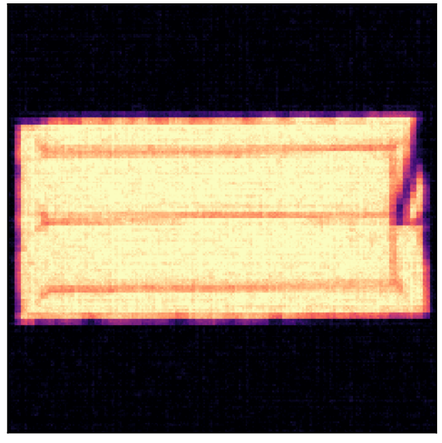
02. CIGS_0158

Frozen validation position 910 | 512 PDHG iterations | PSNR-optimal evaluated rho=0.0001

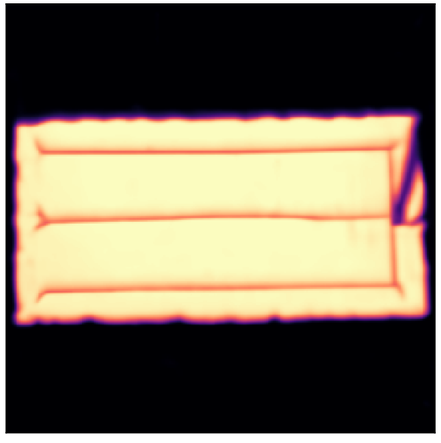
Ground truth



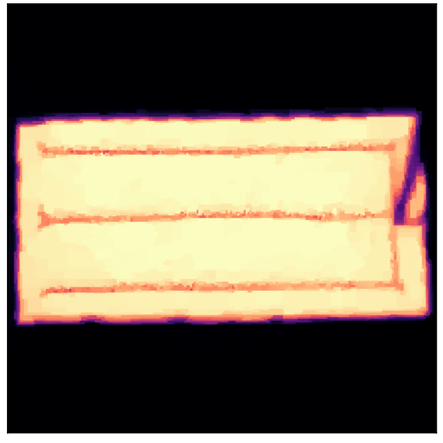
Zero-filled reconstruction
PSNR 25.69 dB | SSIM 0.483
HaarPSI 0.535



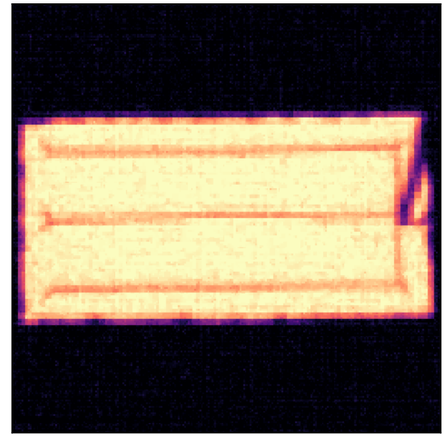
PnP-ADMM-tuned-5
PSNR 40.24 dB | SSIM 0.984
HaarPSI 0.952



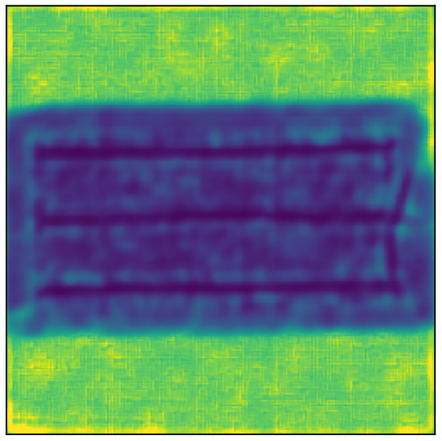
C60 with learned map
PSNR 33.04 dB | SSIM 0.944
HaarPSI 0.825



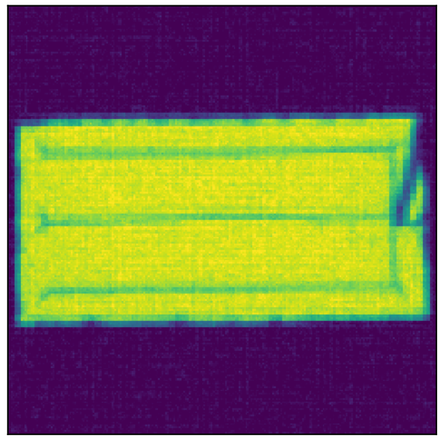
C60 with ZF map (rho=0.0001)
PSNR 25.76 dB | SSIM 0.495
HaarPSI 0.558



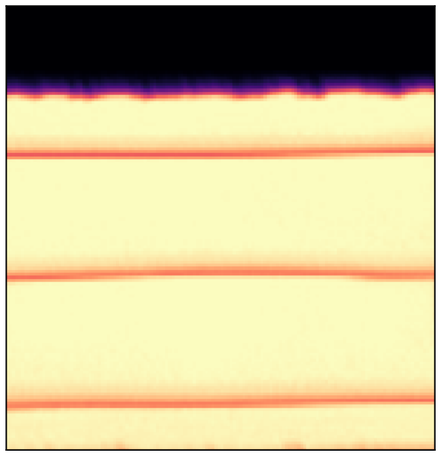
Learned parameter map
min 0.00461 | mean 0.112 | max 0.31



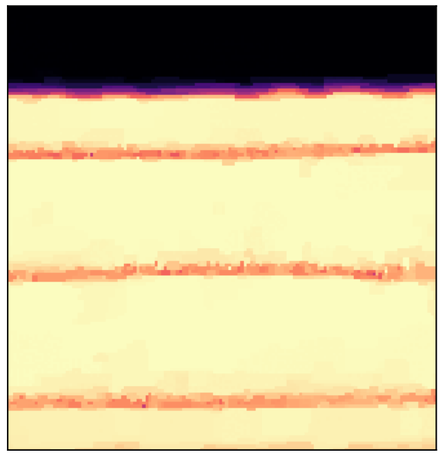
Zero-fill parameter map * rho
min 0 | mean 4.31e-05 | max 0.000112



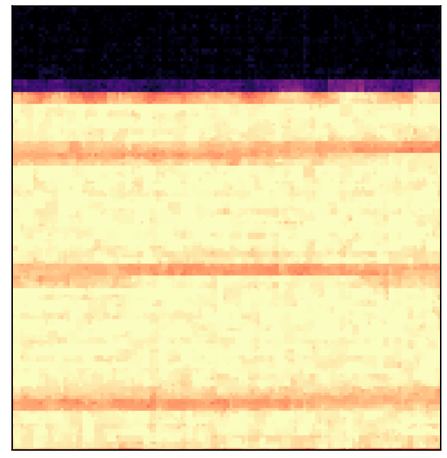
GT texture crop



Learned-map C60 crop



Zero-map C60 crop



PSNR change ZF-map minus learned: -7.280 dB | rho evaluations: 50 | final bracket width: 0.00677

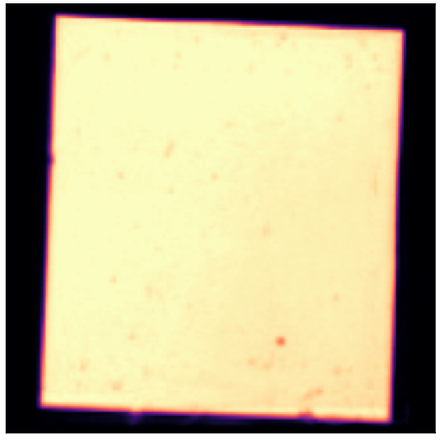
Zero-fill negative pixel fraction: 25.1%. Best rho inside final bracket. Map panels have independent display scales; image panels all use [0,1].

Single-image GT-guided PSNR search. High-quality outputs and search traces are retained in the experiment archive.

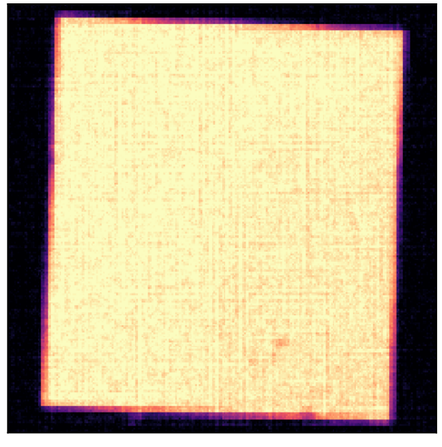
03. Organic_0917

Frozen validation position 902 | 512 PDHG iterations | PSNR-optimal evaluated rho=0.0001

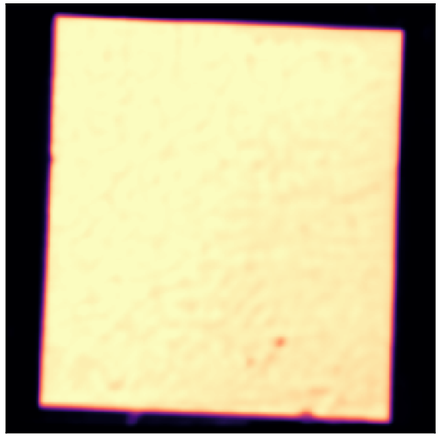
Ground truth



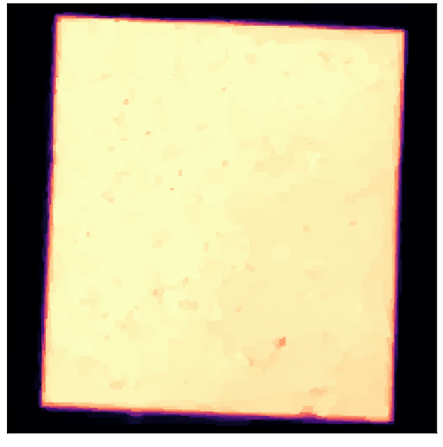
Zero-filled reconstruction
PSNR 25.59 dB | SSIM 0.479
HaarPSI 0.569



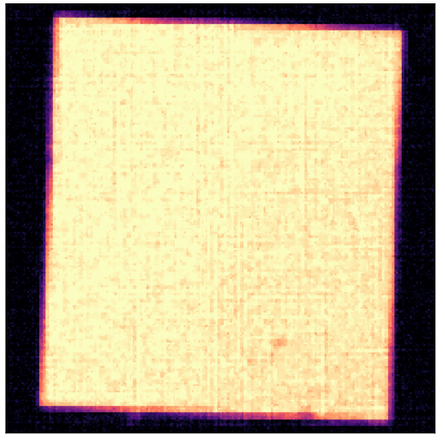
PnP-ADMM-tuned-5
PSNR 39.56 dB | SSIM 0.964
HaarPSI 0.947



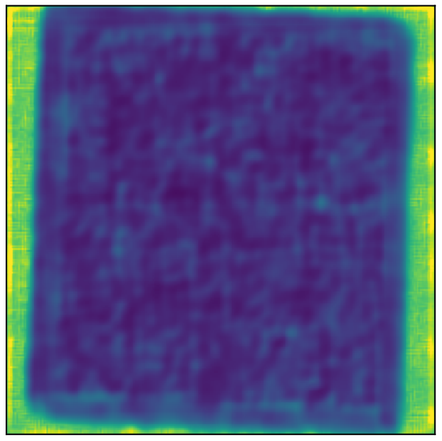
C60 with learned map
PSNR 36.50 dB | SSIM 0.950
HaarPSI 0.900



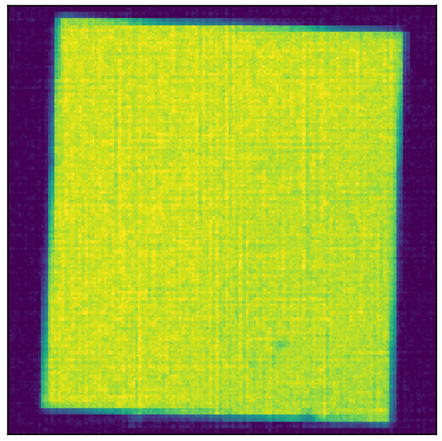
C60 with ZF map (rho=0.0001)
PSNR 25.91 dB | SSIM 0.516
HaarPSI 0.603



Learned parameter map
min 0.00938 | mean 0.0616 | max 0.318



Zero-fill parameter map * rho
min 0 | mean 7.33e-05 | max 0.000113



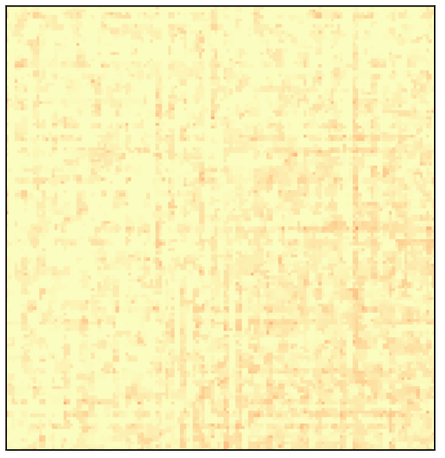
GT texture crop



Learned-map C60 crop



Zero-map C60 crop



PSNR change ZF-map minus learned: -10.589 dB | rho evaluations: 48 | final bracket width: 0.00677

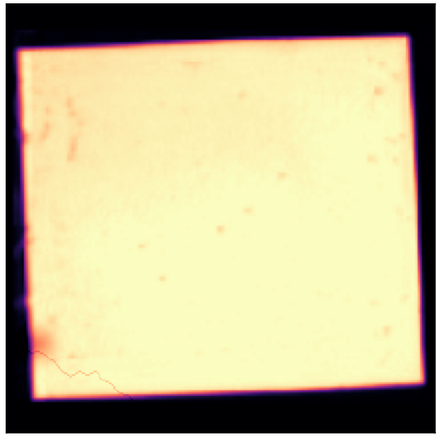
Zero-fill negative pixel fraction: 9.8%. Best rho outside final bracket. Map panels have independent display scales; image panels all use [0,1].

Single-image GT-guided PSNR search. High-quality outputs and search traces are retained in the experiment archive.

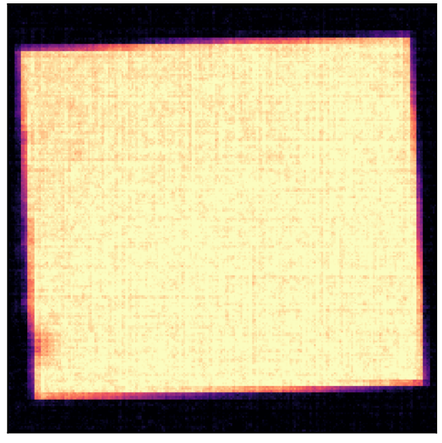
04. Organic_0008

Frozen validation position 903 | 512 PDHG iterations | PSNR-optimal evaluated rho=0.0001

Ground truth



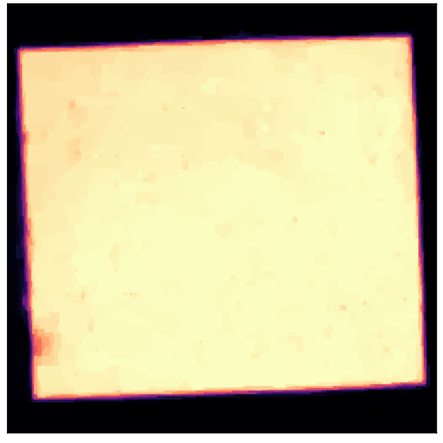
Zero-filled reconstruction
PSNR 25.45 dB | SSIM 0.472
HaarPSI 0.545



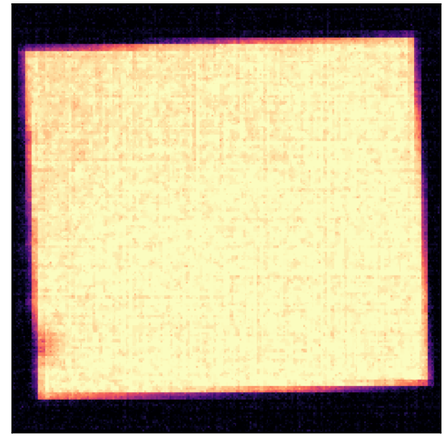
PnP-ADMM-tuned-5
PSNR 39.13 dB | SSIM 0.965
HaarPSI 0.947



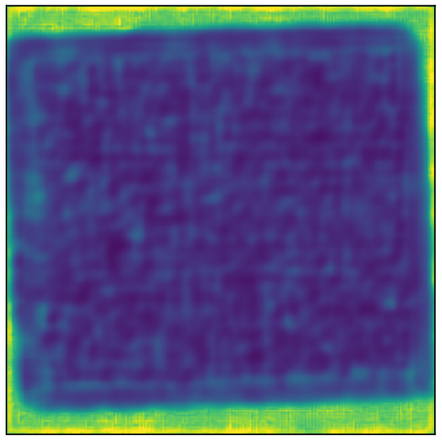
C60 with learned map
PSNR 36.31 dB | SSIM 0.948
HaarPSI 0.888



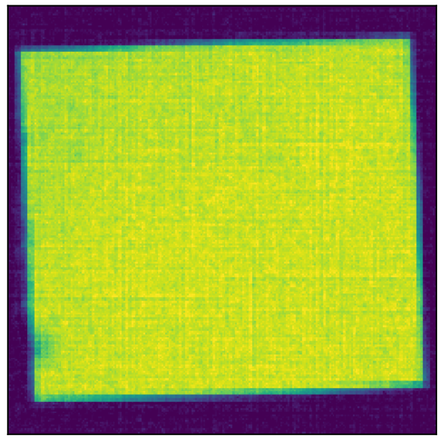
C60 with ZF map (rho=0.0001)
PSNR 25.77 dB | SSIM 0.509
HaarPSI 0.580



Learned parameter map
min 0.0107 | mean 0.0625 | max 0.301



Zero-fill parameter map * rho
min 0 | mean 7.4e-05 | max 0.000114



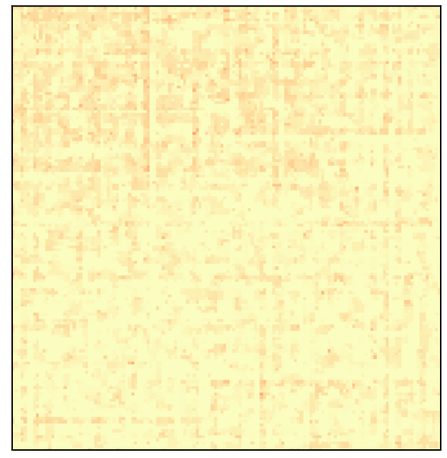
GT texture crop



Learned-map C60 crop



Zero-map C60 crop



PSNR change ZF-map minus learned: -10.534 dB | rho evaluations: 48 | final bracket width: 0.00677

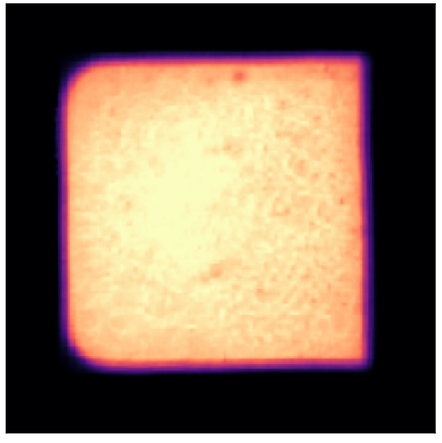
Zero-fill negative pixel fraction: 9.5%. Best rho outside final bracket. Map panels have independent display scales; image panels all use [0,1].

Single-image GT-guided PSNR search. High-quality outputs and search traces are retained in the experiment archive.

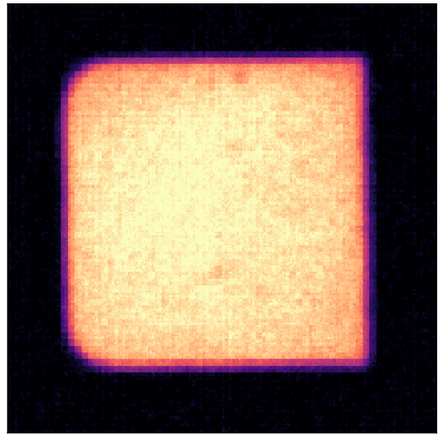
05. Perovskite_0733

Frozen validation position 901 | 512 PDHG iterations | PSNR-optimal evaluated rho=0.0001

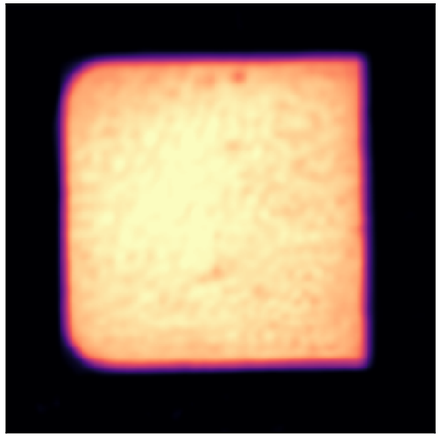
Ground truth



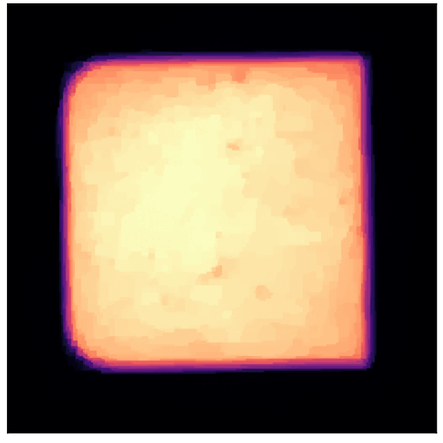
Zero-filled reconstruction
PSNR 28.88 dB | SSIM 0.543
HaarPSI 0.685



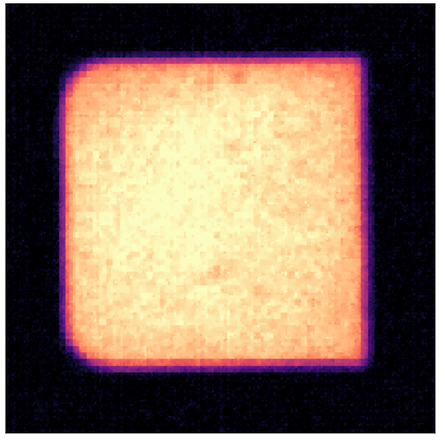
PnP-ADMM-tuned-5
PSNR 38.56 dB | SSIM 0.946
HaarPSI 0.922



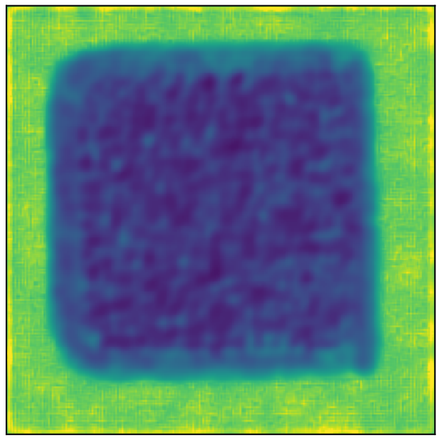
C60 with learned map
PSNR 36.29 dB | SSIM 0.890
HaarPSI 0.854



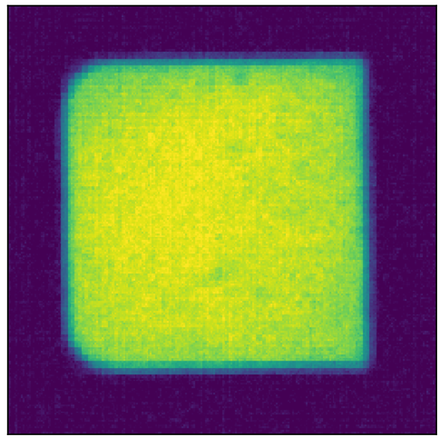
C60 with ZF map (rho=0.0001)
PSNR 28.71 dB | SSIM 0.559
HaarPSI 0.706



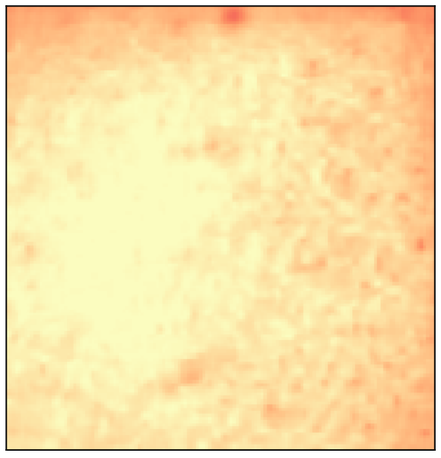
Learned parameter map
min 0.0138 | mean 0.107 | max 0.294



Zero-fill parameter map * rho
min 0 | mean 4.63e-05 | max 0.00011



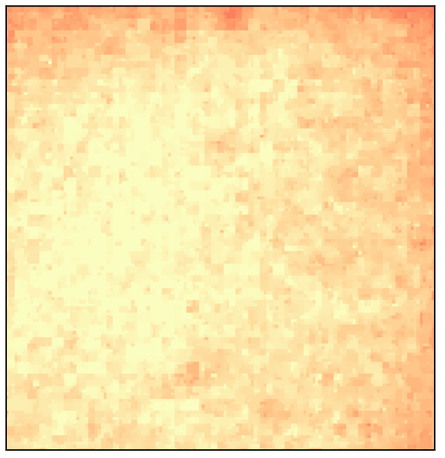
GT texture crop



Learned-map C60 crop



Zero-map C60 crop



PSNR change ZF-map minus learned: -7.584 dB | rho evaluations: 50 | final bracket width: 0.00677

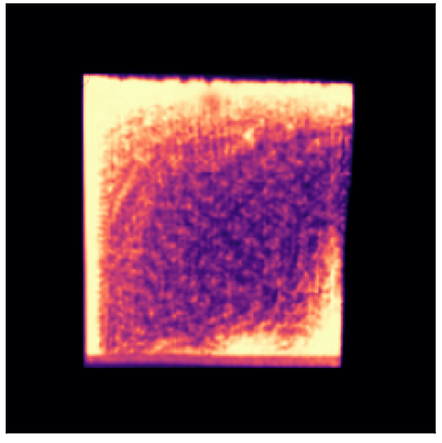
Zero-fill negative pixel fraction: 21.6%. Best rho inside final bracket. Map panels have independent display scales; image panels all use [0,1].

Single-image GT-guided PSNR search. High-quality outputs and search traces are retained in the experiment archive.

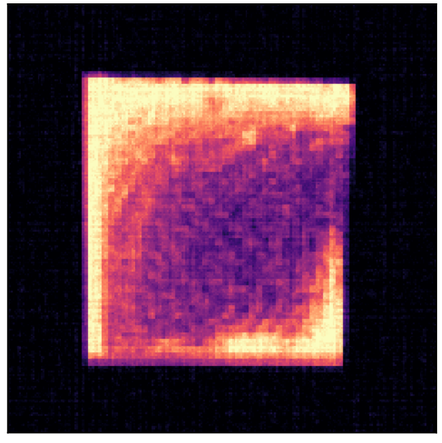
06. Perovskite_0396

Frozen validation position 906 | 512 PDHG iterations | PSNR-optimal evaluated rho=0.0001

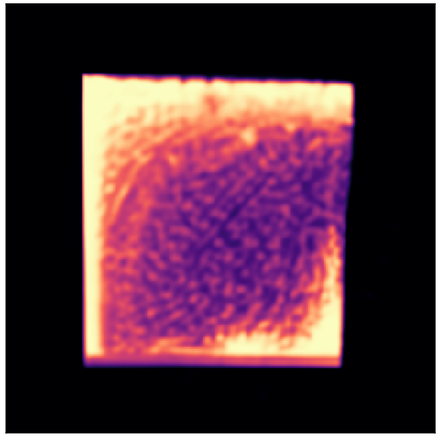
Ground truth



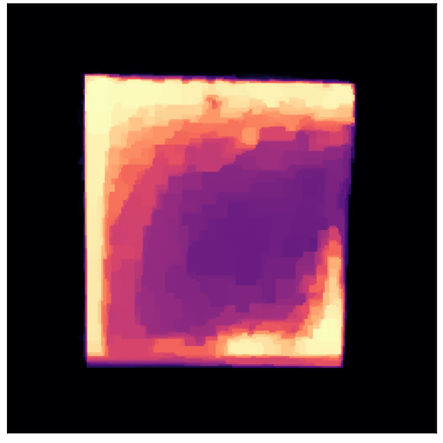
Zero-filled reconstruction
PSNR 24.60 dB | SSIM 0.473
HaarPSI 0.515



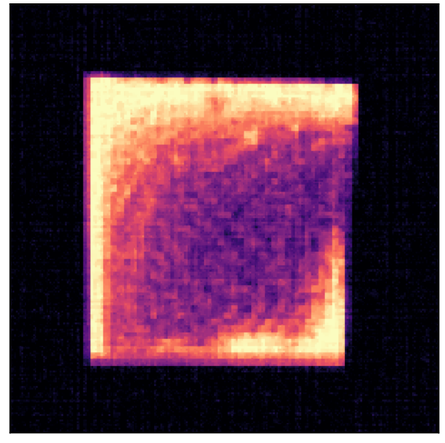
PnP-ADMM-tuned-5
PSNR 32.52 dB | SSIM 0.931
HaarPSI 0.818



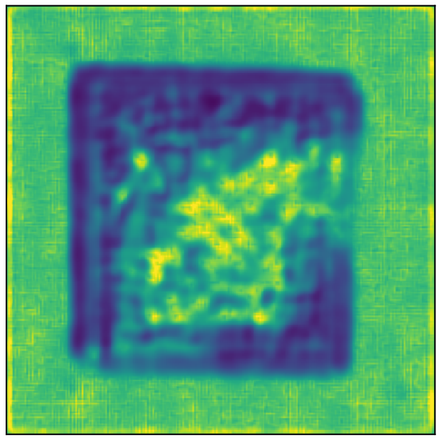
C60 with learned map
PSNR 28.56 dB | SSIM 0.803
HaarPSI 0.552



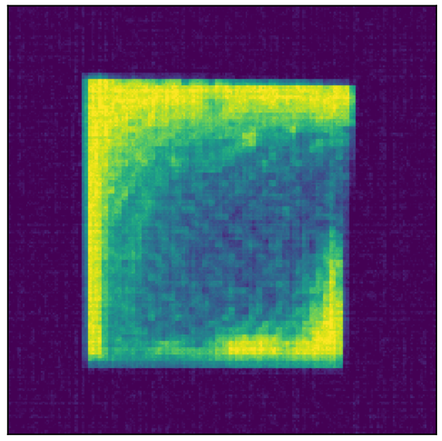
C60 with ZF map (rho=0.0001)
PSNR 24.71 dB | SSIM 0.467
HaarPSI 0.524



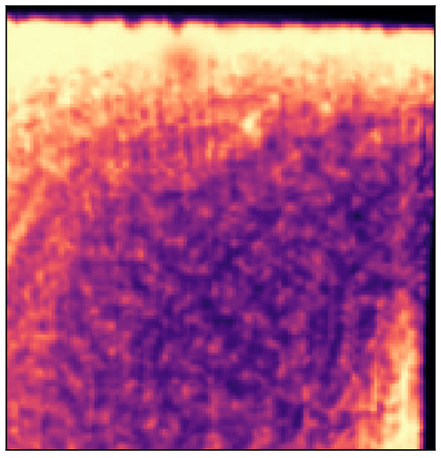
Learned parameter map
min 0.00891 | mean 0.15 | max 0.349



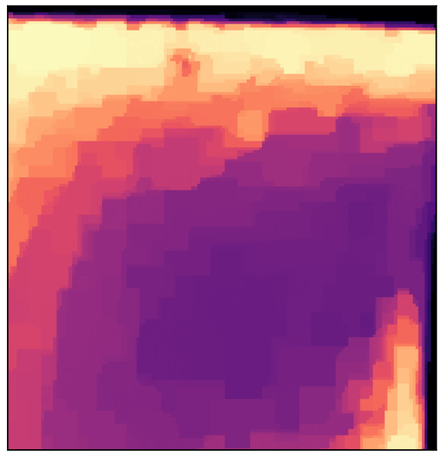
Zero-fill parameter map * rho
min 0 | mean 2.57e-05 | max 0.000112



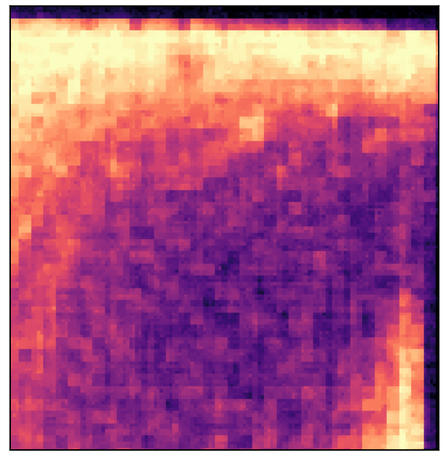
GT texture crop



Learned-map C60 crop



Zero-map C60 crop



PSNR change ZF-map minus learned: -3.849 dB | rho evaluations: 50 | final bracket width: 0.00677

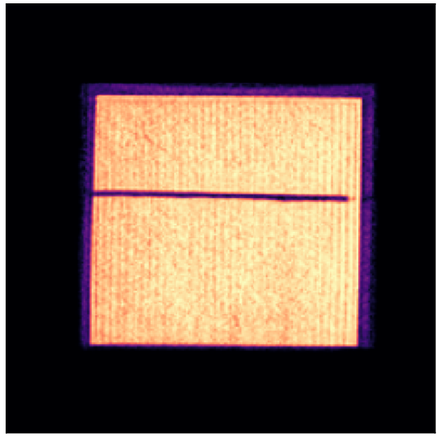
Zero-fill negative pixel fraction: 28.2%. Best rho inside final bracket. Map panels have independent display scales; image panels all use [0,1].

Single-image GT-guided PSNR search. High-quality outputs and search traces are retained in the experiment archive.

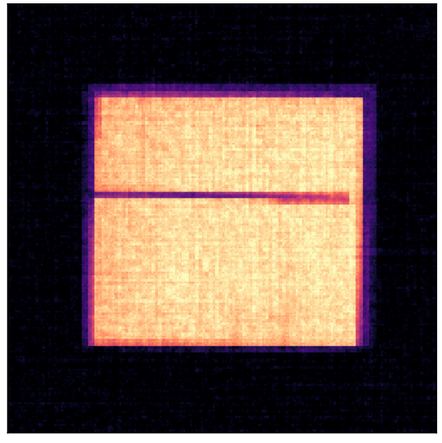
07. Reference_0768

Frozen validation position 907 | 512 PDHG iterations | PSNR-optimal evaluated rho=0.0001

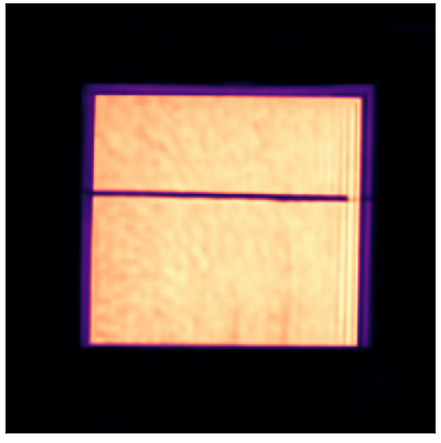
Ground truth



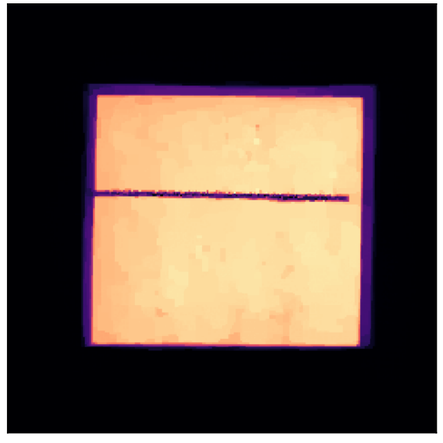
Zero-filled reconstruction
PSNR 25.09 dB | SSIM 0.436
HaarPSI 0.531



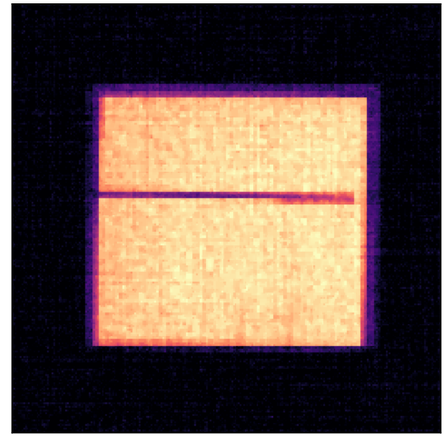
PnP-ADMM-tuned-5
PSNR 33.48 dB | SSIM 0.860
HaarPSI 0.789



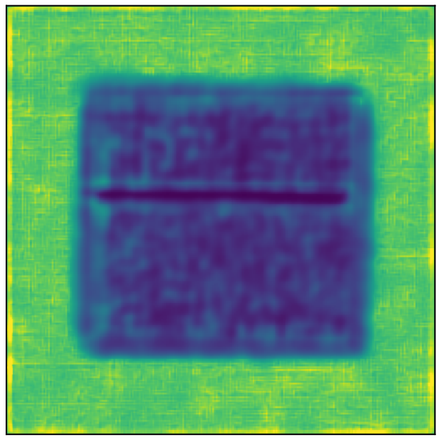
C60 with learned map
PSNR 31.75 dB | SSIM 0.807
HaarPSI 0.666



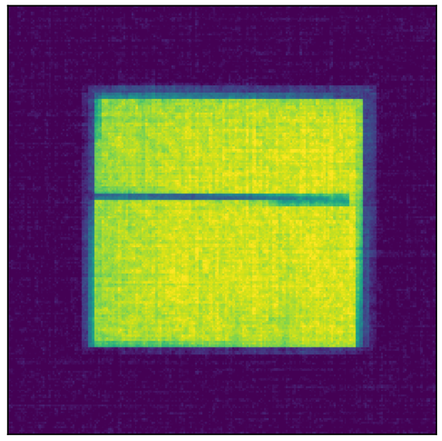
C60 with ZF map (rho=0.0001)
PSNR 25.09 dB | SSIM 0.420
HaarPSI 0.537



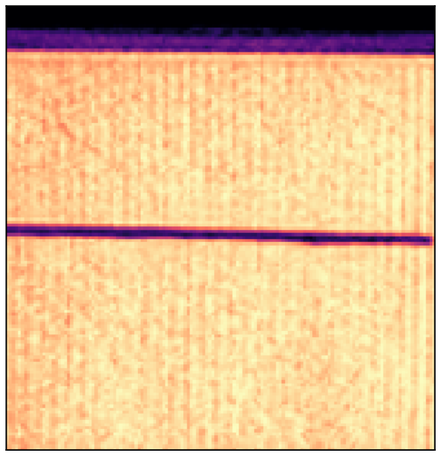
Learned parameter map
min 0.00289 | mean 0.128 | max 0.319



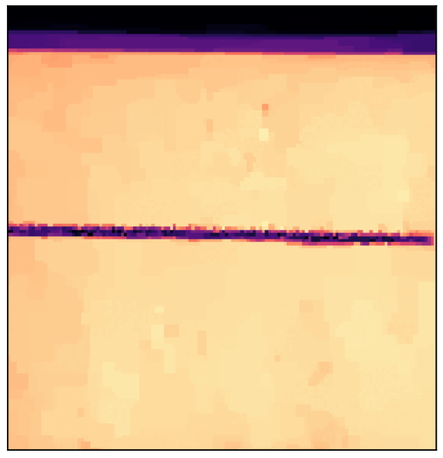
Zero-fill parameter map * rho
min 0 | mean 3.44e-05 | max 0.000106



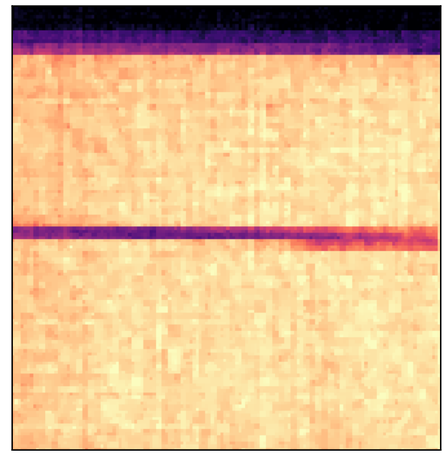
GT texture crop



Learned-map C60 crop



Zero-map C60 crop



PSNR change ZF-map minus learned: -6.665 dB | rho evaluations: 50 | final bracket width: 0.00677

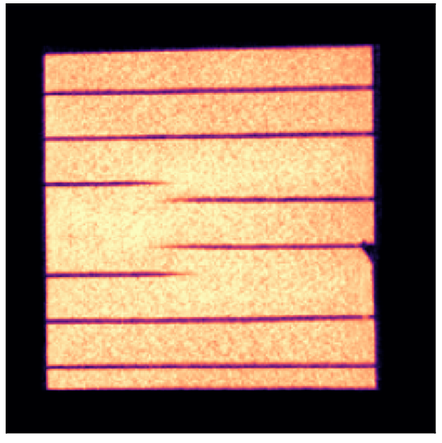
Zero-fill negative pixel fraction: 28.1%. Best rho inside final bracket. Map panels have independent display scales; image panels all use [0,1].

Single-image GT-guided PSNR search. High-quality outputs and search traces are retained in the experiment archive.

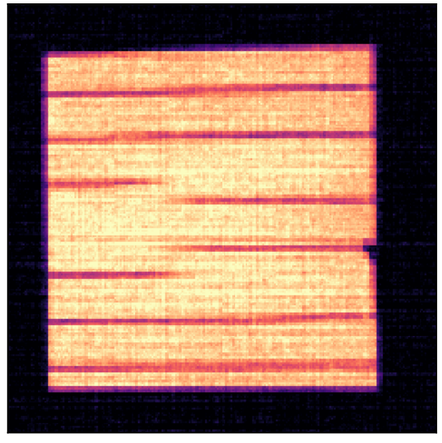
08. Reference_0589

Frozen validation position 909 | 512 PDHG iterations | PSNR-optimal evaluated rho=0.0001

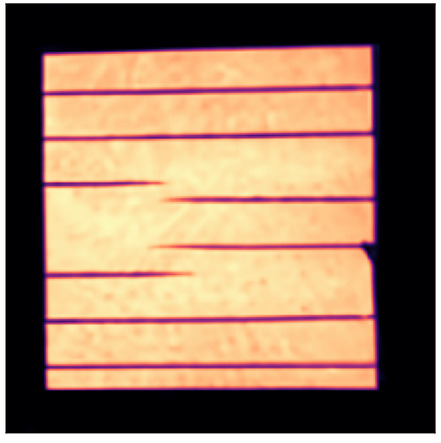
Ground truth



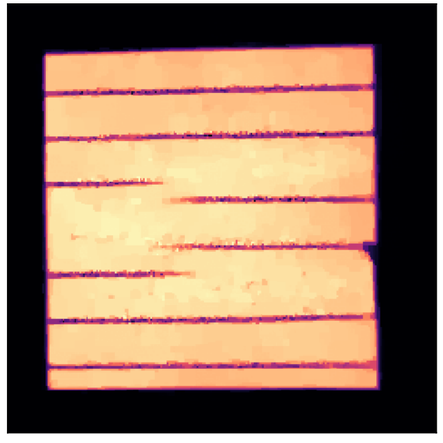
Zero-filled reconstruction
PSNR 21.63 dB | SSIM 0.422
HaarPSI 0.484



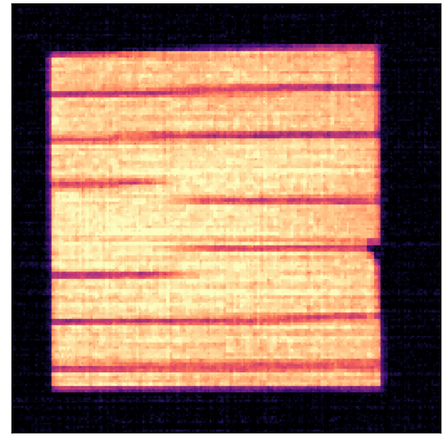
PnP-ADMM-tuned-5
PSNR 32.07 dB | SSIM 0.840
HaarPSI 0.799



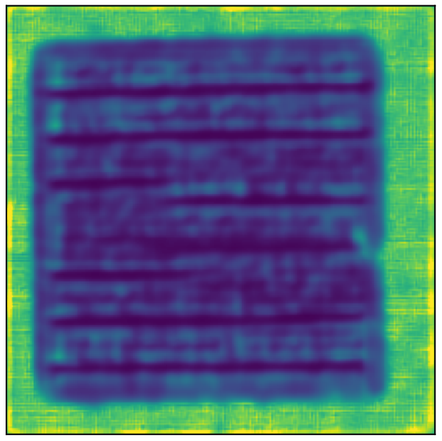
C60 with learned map
PSNR 28.39 dB | SSIM 0.798
HaarPSI 0.723



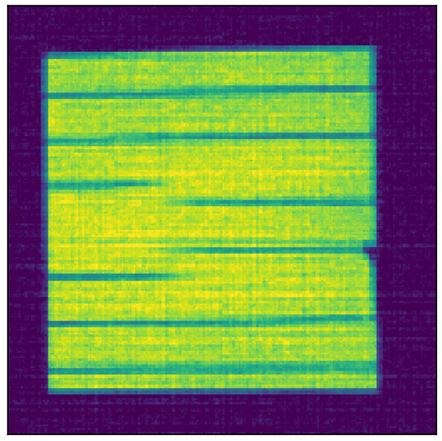
C60 with ZF map (rho=0.0001)
PSNR 21.81 dB | SSIM 0.426
HaarPSI 0.496



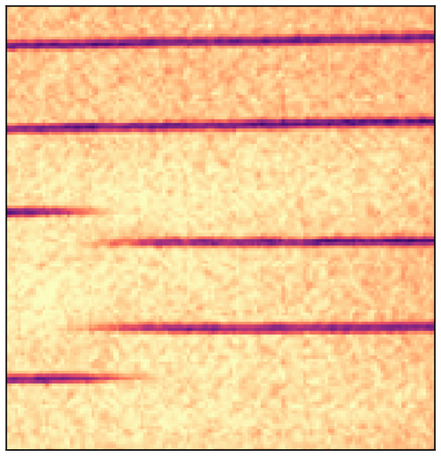
Learned parameter map
min 0.00223 | mean 0.0913 | max 0.365



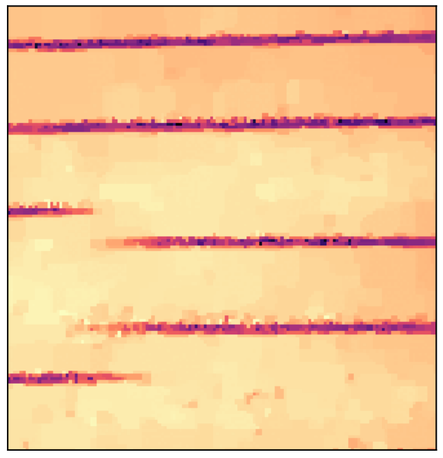
Zero-fill parameter map * rho
min 0 | mean 5.3e-05 | max 0.000115



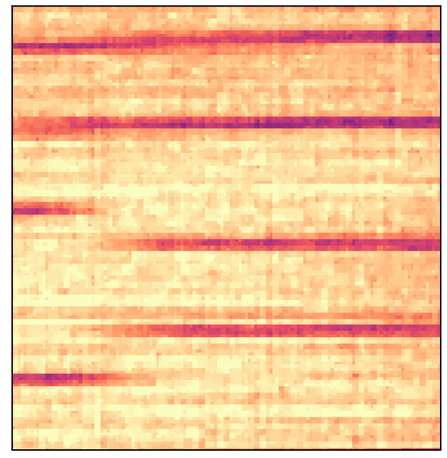
GT texture crop



Learned-map C60 crop



Zero-map C60 crop



PSNR change ZF-map minus learned: -6.575 dB | rho evaluations: 50 | final bracket width: 0.00677

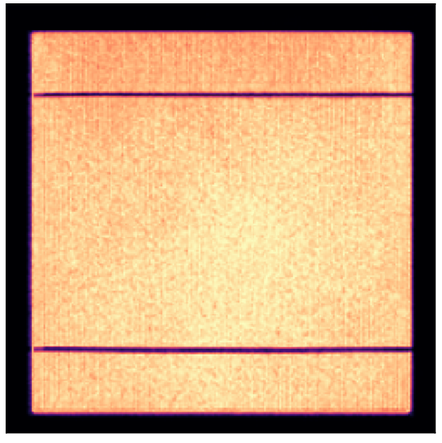
Zero-fill negative pixel fraction: 17.9%. Best rho inside final bracket. Map panels have independent display scales; image panels all use [0,1].

Single-image GT-guided PSNR search. High-quality outputs and search traces are retained in the experiment archive.

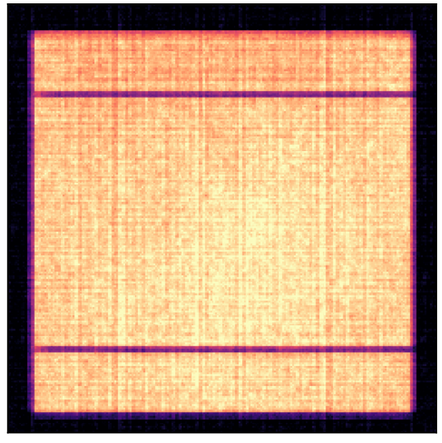
09. Silicone_0008

Frozen validation position 903 | 512 PDHG iterations | PSNR-optimal evaluated rho=0.001

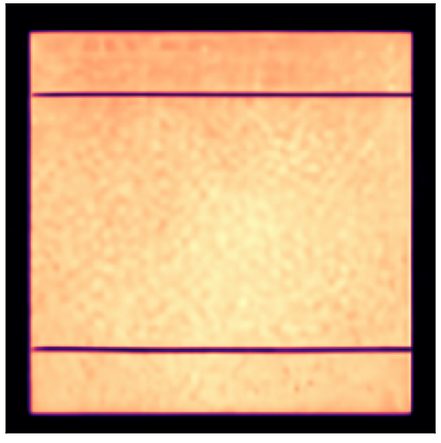
Ground truth



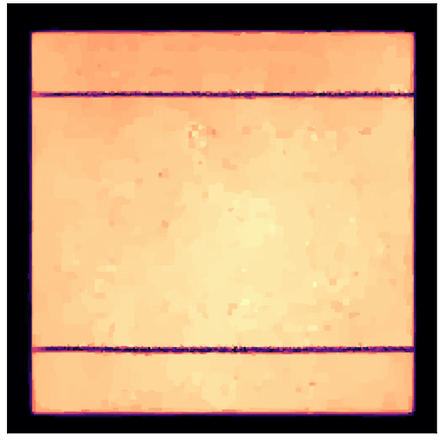
Zero-filled reconstruction
PSNR 22.69 dB | SSIM 0.427
HaarPSI 0.533



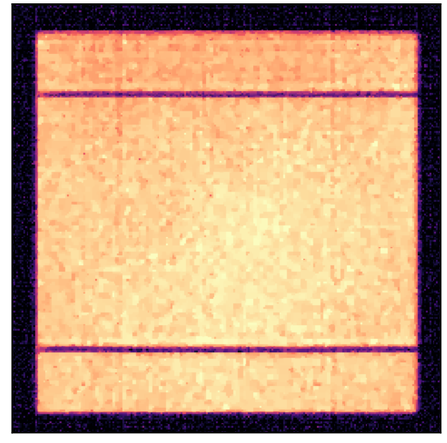
PnP-ADMM-tuned-5
PSNR 31.87 dB | SSIM 0.743
HaarPSI 0.843



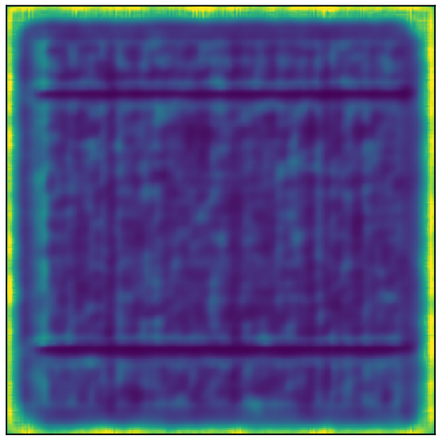
C60 with learned map
PSNR 29.79 dB | SSIM 0.718
HaarPSI 0.803



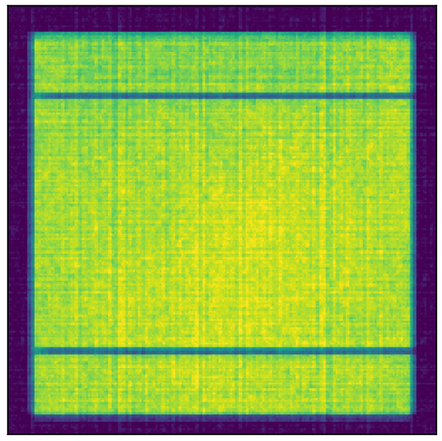
C60 with ZF map (rho=0.001)
PSNR 23.25 dB | SSIM 0.533
HaarPSI 0.574



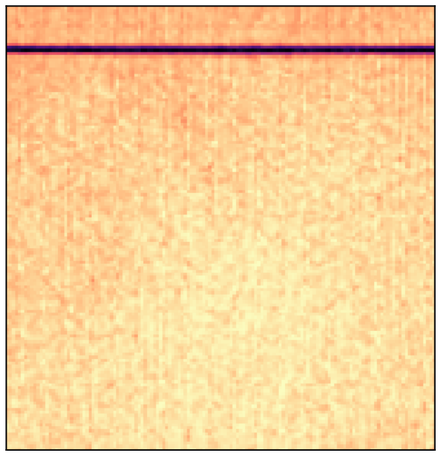
Learned parameter map
min 0.00184 | mean 0.0614 | max 0.379



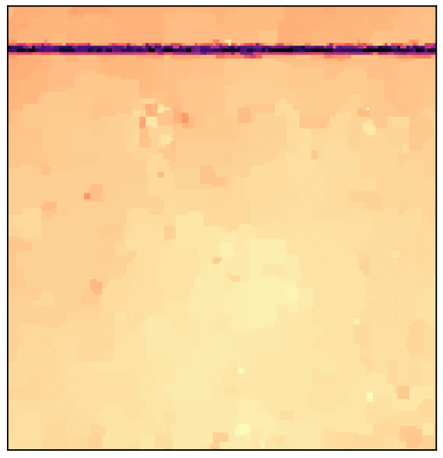
Zero-fill parameter map * rho
min 0 | mean 0.000703 | max 0.00113



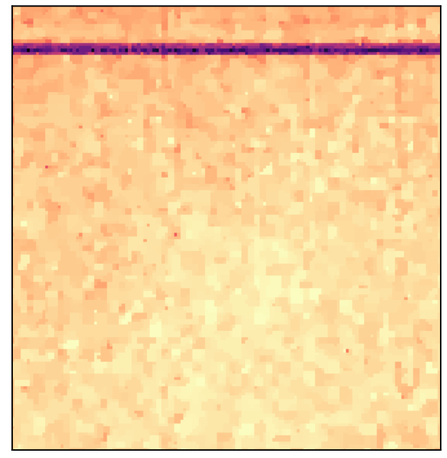
GT texture crop



Learned-map C60 crop



Zero-map C60 crop



PSNR change ZF-map minus learned: -6.542 dB | rho evaluations: 48 | final bracket width: 0.00677

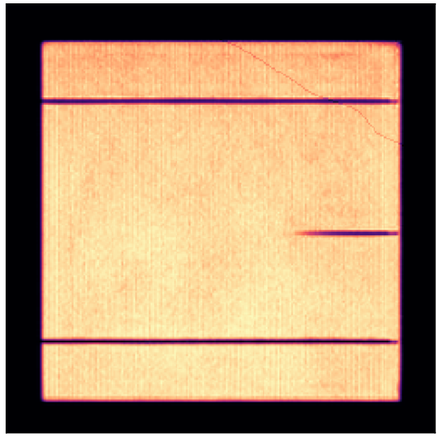
Zero-fill negative pixel fraction: 9.0%. Best rho outside final bracket. Map panels have independent display scales; image panels all use [0,1].

Single-image GT-guided PSNR search. High-quality outputs and search traces are retained in the experiment archive.

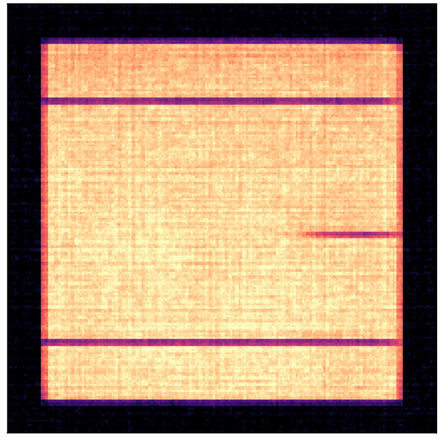
10. Silicone_0589

Frozen validation position 909 | 512 PDHG iterations | PSNR-optimal evaluated rho=0.0001

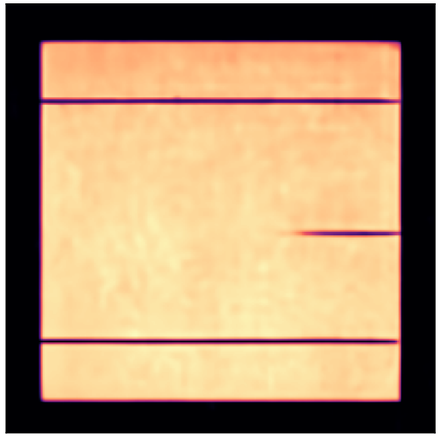
Ground truth



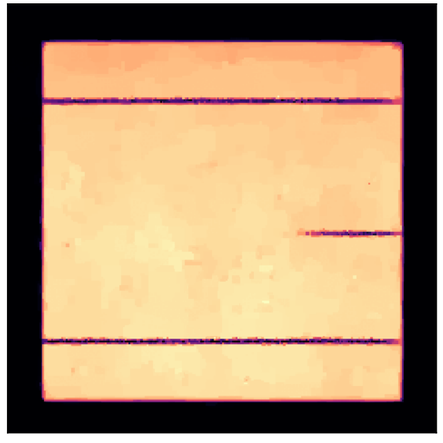
Zero-filled reconstruction
PSNR 22.33 dB | SSIM 0.437
HaarPSI 0.465



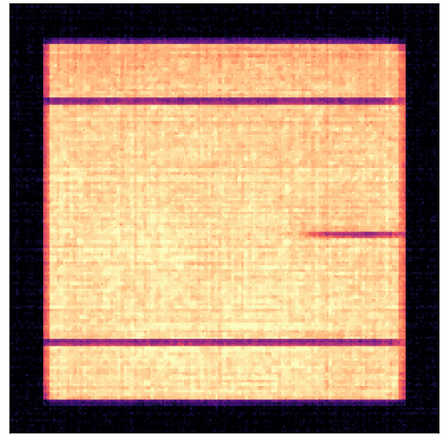
PnP-ADMM-tuned-5
PSNR 34.15 dB | SSIM 0.833
HaarPSI 0.851



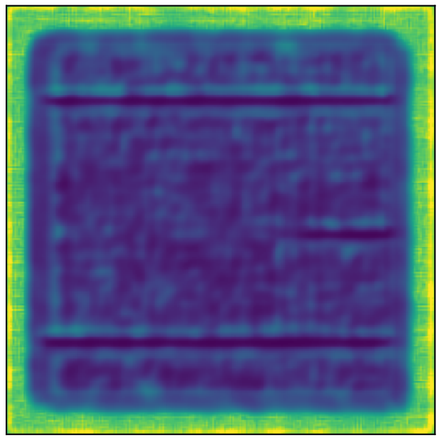
C60 with learned map
PSNR 31.43 dB | SSIM 0.818
HaarPSI 0.772



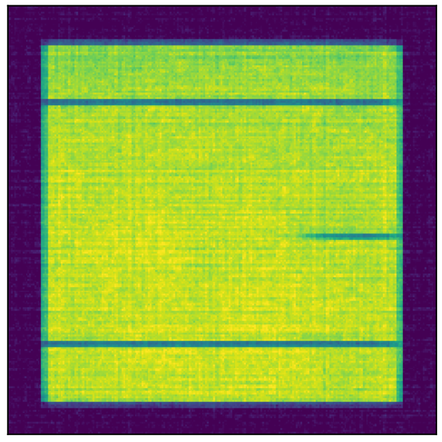
C60 with ZF map (rho=0.0001)
PSNR 22.55 dB | SSIM 0.450
HaarPSI 0.479



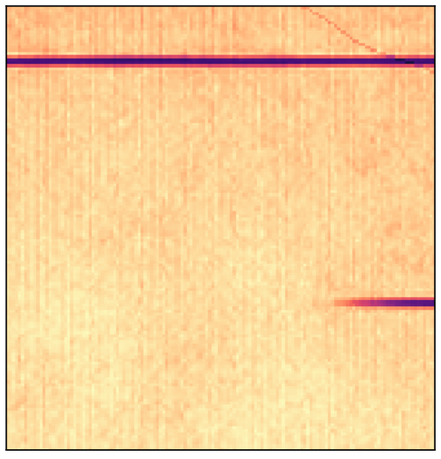
Learned parameter map
min 0.00257 | mean 0.0759 | max 0.287



Zero-fill parameter map * rho
min 0 | mean 6.3e-05 | max 0.000111



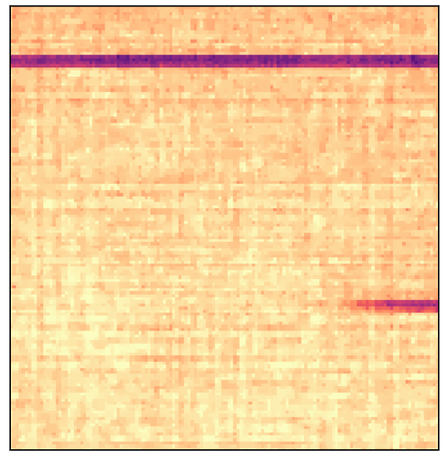
GT texture crop



Learned-map C60 crop



Zero-map C60 crop



PSNR change ZF-map minus learned: -8.876 dB | rho evaluations: 50 | final bracket width: 0.00677

Zero-fill negative pixel fraction: 13.5%. Best rho inside final bracket. Map panels have independent display scales; image panels all use [0,1].

Single-image GT-guided PSNR search. High-quality outputs and search traces are retained in the experiment archive.